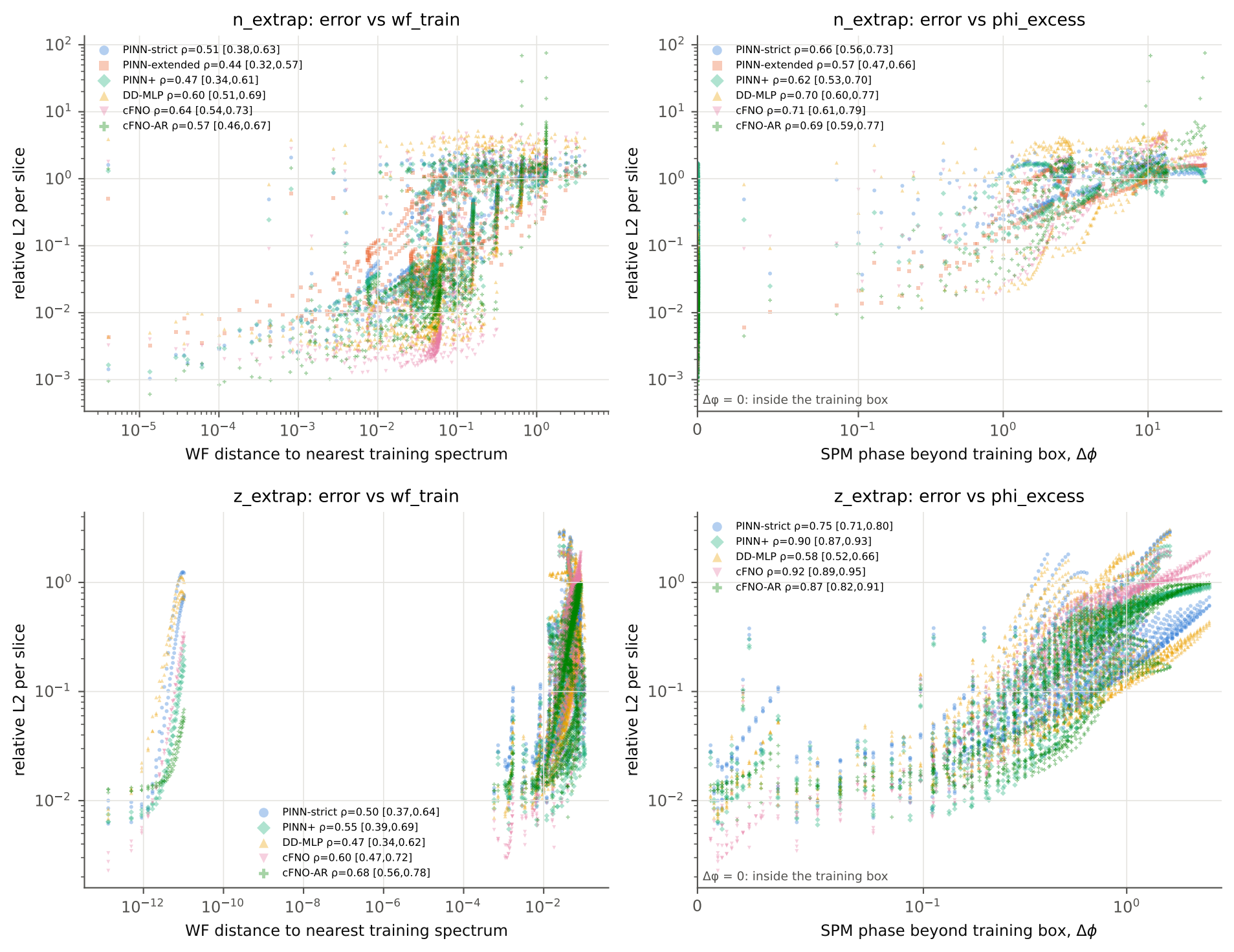

n_extrap: error vs wf_train
PINN-strict ρ=0.51 [0.38,0.63]
PINN-extended ρ=0.44 [0.32,0.57]
PINN+ ρ=0.47 [0.34,0.61]
DD-MLP ρ=0.60 [0.51,0.69]
cFNO ρ=0.64 [0.54,0.73]
cFNO-AR ρ=0.57 [0.46,0.67]
relative L2 per slice
WF distance to nearest training spectrum
n_extrap: error vs phi_excess
PINN-strict ρ=0.66 [0.56,0.73]
PINN-extended ρ=0.57 [0.47,0.66]
PINN+ ρ=0.62 [0.53,0.70]
DD-MLP ρ=0.70 [0.60,0.77]
cFNO ρ=0.71 [0.61,0.79]
cFNO-AR ρ=0.69 [0.59,0.77]
Δφ = 0: inside the training box
SPM phase beyond training box, Δϕ
z_extrap: error vs wf_train
PINN-strict ρ=0.50 [0.37,0.64]
PINN+ ρ=0.55 [0.39,0.69]
DD-MLP ρ=0.47 [0.34,0.62]
cFNO ρ=0.60 [0.47,0.72]
cFNO-AR ρ=0.68 [0.56,0.78]
relative L2 per slice
WF distance to nearest training spectrum
z_extrap: error vs phi_excess
PINN-strict ρ=0.75 [0.71,0.80]
PINN+ ρ=0.90 [0.87,0.93]
DD-MLP ρ=0.58 [0.52,0.66]
cFNO ρ=0.92 [0.89,0.95]
cFNO-AR ρ=0.87 [0.82,0.91]
Δφ = 0: inside the training box
SPM phase beyond training box, Δϕ